Pico W network meter

Shows how crowded each 2.4 GHz Wi-Fi channel is, so you can pick the best channel for your router.

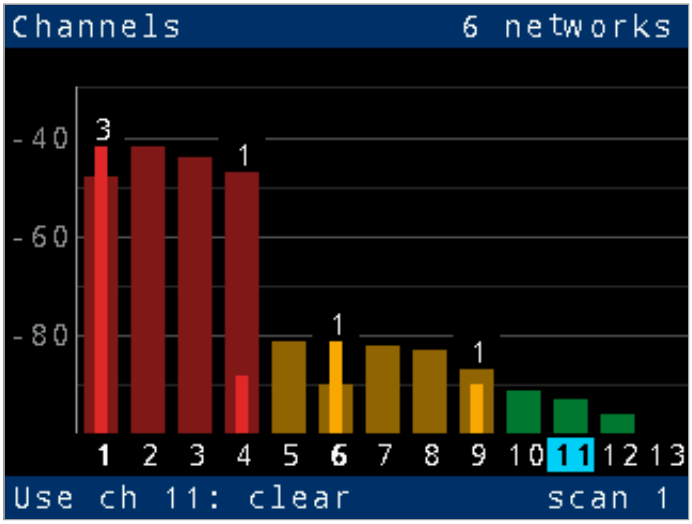
Buttons

A	Next screen
B	Previous screen
X	Scan now
Y	Not used

LED

Blue	Scanning
Green	Recommended channel is clear
Amber	Recommended channel is moderately busy
Red	Even the best channel is busy
Magenta	Last scan failed (retried in 2 s; restarts after 5 in a row)

Channels screen



- **Bright bar**: signal, the strongest network on the channel.
- **Dim bar**: interference, every other network on the channel or up to 3 channels away.
- **Scale**: dBm, with a grid line every 10 dB.
- **Colour**: how busy, counting both: green clear, amber moderate, red busy.
- **Number above a bar**: networks on that channel.
- **Bold channel numbers**: the standard channels 1, 6 and 11, which don't overlap.
- **Highlighted channel**: the one to use, the quietest of 1, 6 and 11. The footer names it: "Use ch 11: clear".

Networks screen



- Strongest first: **name**, **channel**, **signal** in dBm (closer to 0 is stronger).
- **Signal colour**: green good (≥ −60), amber fair (≥ −75), red weak.
- Names over 24 characters are shortened in the middle, keeping the end.
- Names in **amber** are open networks (no password).
- Names in **grey** are hidden networks, shown by the end of their MAC address: "hidden a2:7f:06".

Fading

A network stays at full brightness for **15 s** after it was last heard, then fades and is gone by **30 s**. Dim bars and rows are old data. A network that is heard again is bright again straight away.

Good to know

- Scans run every 10 s. Press X after moving to a new spot.
- Measure where you use Wi-Fi, not next to the router.
- 2.4 GHz only: the Pico W cannot hear 5 GHz networks.
- Keep it away from USB 3 hubs: their noise drowns out all but the strongest networks. Use a long cable or a USB charger.
- Serial log: over the Pico's USB socket, `/dev/cu.usbmodem…` on a Mac (`screen /dev/cu.usbmodem<N> 115200`).